

TEST DESIGN DOCUMENT

(TDD)

Legal Contract Q&A Assistant using Generative AI

Professional Project Documentation

Document Type	Test Design Document
Project	Legal Contract Q&A Assistant
AI Engine	Gemini AI (RAG Pipeline)
Version	1.0

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1. Introduction

1.1 Purpose

This Test Design Document (TDD) defines the testing approach, scope, and criteria for the Legal Contract Q&A Assistant — an AI-powered application that allows users to upload legal contracts and ask natural-language questions, answered through a Retrieval-Augmented Generation (RAG) pipeline powered by Gemini AI.

1.2 Project Overview

The system enables users to securely upload contracts, have the text processed and indexed, and receive accurate, context-aware answers to questions about the contract's content. The document establishes what will be tested, how, by whom, and under what conditions the testing effort is considered complete.

2. Objectives

- Reduce manual contract review time through AI-assisted analysis.
- Improve legal document accessibility for non-legal users.
- Provide accurate, context-aware AI-assisted responses.
- Enhance productivity through intelligent semantic search.
- Validate the reliability and security of the end-to-end pipeline.

3. Scope

3.1 In Scope

- Functional testing of all modules listed in Section 4.
- Non-functional testing: performance, security, usability, compatibility, and reliability.
- Integration testing of the RAG pipeline (chunking → embeddings → FAISS → Gemini).

3.2 Out of Scope

- Testing of third-party infrastructure internals (Supabase, Gemini API, FAISS library internals).
- Load testing beyond expected production traffic volumes.

4. Modules Under Testing

- User Registration & Login (Supabase Authentication)
- Dashboard
- Contract Upload
- Document Processing
- Text Chunking
- Embedding Generation

- FAISS Vector Database
- RAG Retrieval Pipeline
- Gemini AI Response Generation
- Chat Interface

5. Technology Stack

Component	Technology
Frontend	React
Backend	Python
Authentication & Database	Supabase
Embeddings	Sentence Transformers
Vector Database	FAISS
Generative AI	Gemini
Dataset	CUAD (Contract Understanding Atticus Dataset)

6. Test Levels & Responsibilities

Test Level	Owner	Description
Unit Testing	Developers	Validate individual functions/components (e.g., chunking logic, embedding calls) in isolation.
Integration Testing	QA / Developers	Verify interaction between modules — e.g., upload → processing → chunking → embeddings → FAISS.
System Testing	QA Team	Validate the end-to-end flow from login through contract upload to AI-generated answers.
User Acceptance Testing (UAT)	Business / End Users	Confirm the system meets business requirements and is usable by legal/non-technical staff.

7. Functional Test Cases

The table below lists representative functional test cases covering the core user journeys and pipeline stages. Each test case should be logged with actual results, evidence, and defect references during execution.

Test ID	Module	Test Scenario	Expected Result	Status
TC-01	Registration	Register with valid details	Account is created and user is redirected to login	Pass
TC-02	Registration	Register with an email already in use	System displays a duplicate-account error	Pass
TC-03	Login	Login with valid credentials	User is authenticated and dashboard is displayed	Pass
TC-04	Login	Login with invalid credentials	System displays an authentication error	Pass
TC-05	Dashboard	Load dashboard after login	Dashboard renders with uploaded contracts and options	Pass
TC-06	Upload	Upload a valid PDF contract	File uploads successfully and appears in contract list	Pass
TC-07	Upload	Upload an unsupported file type	System rejects file with a clear error message	Pass
TC-08	Processing	Extract text from an uploaded contract	Full contract text is extracted without corruption	Pass
TC-09	Chunking	Split extracted text into chunks	Text is divided into coherent, appropriately sized chunks	Pass
TC-10	Embeddings	Generate embeddings for chunks	Embedding vectors are generated for every chunk	Pass
TC-11	FAISS	Store embeddings in the vector database	FAISS index is created and populated correctly	Pass

Test ID	Module	Test Scenario	Expected Result	Status
TC-12	RAG Retrieval	Ask a question about contract content	Most relevant chunks are retrieved from the index	In Progress
TC-13	Gemini AI	Generate a response using retrieved context	Response is accurate, relevant, and context-aware	In Progress
TC-14	Chat Interface	Send and receive multiple chat messages	Conversation history is displayed correctly	In Progress
TC-15	Security	Attempt to access another user's contract	Access is denied and action is logged	In Progress

8. Non-Functional Testing

8.1 Performance Testing

Validate response times for document upload, processing, and AI-generated answers under expected user load.

8.2 Security Testing

Verify authentication, authorization, data encryption, and protection against unauthorized access to uploaded contracts.

8.3 Usability Testing

Assess ease of navigation, clarity of the chat interface, and overall user experience for non-technical users.

8.4 Compatibility Testing

Confirm consistent behavior across major browsers, screen sizes, and operating systems.

8.5 Reliability Testing

Verify system stability and graceful error handling during extended use and intermittent connectivity.

9. Entry Criteria

- Development is completed for the sprint/release under test.
- Supabase authentication and database are configured.
- Backend APIs are available and accessible in the test environment.
- Test environment and test data are prepared and ready.

10. Exit Criteria

- All critical and high-priority test cases have been executed.
- Authentication and authorization flows are fully validated.
- The RAG pipeline (retrieval and response generation) is verified end-to-end.
- No high-severity or critical defects remain open.

11. Test Environment

Testing will be performed in a dedicated staging environment that mirrors production configuration, including Supabase authentication, FAISS vector storage, and access to the Gemini API under test-tier credentials.

12. Risks & Mitigation

Risk	Impact	Mitigation
Large contracts may increase processing time.	Medium	Implement asynchronous processing and chunk-size optimization.
AI response quality depends on retrieved context.	High	Tune retrieval parameters and evaluate with a labeled QA dataset.
Internet connectivity is required for Gemini API.	Medium	Provide clear offline-state handling and retry logic.
Sensitive legal data exposure.	High	Enforce encryption at rest/in transit and strict access controls.

13. Roles & Responsibilities

Role	Responsibility
Test Lead	Define test strategy, review test cases, sign off on releases.
QA Engineer	Design and execute functional, integration, and regression tests.
Developer	Perform unit testing and fix defects raised during test cycles.

Role	Responsibility
Product Owner	Validate business requirements and approve UAT results.

14. Deliverables

- Test Cases
- Bug Reports
- Test Execution Report
- Test Summary Report
- Final Validation Report

15. Conclusion

This testing strategy ensures that the Legal Contract Q&A Assistant delivers a secure, reliable, and efficient user experience while meeting all functional and business requirements. Execution results, defects, and sign-off should be tracked against this document throughout the test cycle.